

Janani Pradeep

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EDUCATION

Master of Computer Science

May 2025

North Carolina State University, Raleigh, NC

GPA: 4.0/4.0

Coursework: DBMS, Automated Learning & Data Analysis, Design & Analysis of Algorithms, Computer Networks, Statistics, Neural Networks & Deep Learning, Cloud Computing, Social Computing & Decentralized AI, Privacy in Digital Age, Applied Artificial Intelligence

Bachelor of Technology in Computer Science Engineering

May 2022

Jaypee Institute of Information Technology, Noida, India

CGPA: 9.6/10

Coursework: Big Data Analytics with Hadoop and Spark, Database and Automata, Agile Software Development, Software Engineering, Computer Networks, Operating System, Computer Architecture, Project Management

TECHNICAL SKILLS

Languages: Python, PHP, C, C++, HTML5, CSS3, JS, R Programming, VBA, AppScript, Xamarin, ReactJS, Typescript

Databases and Visualization Tools: Microsoft SQL, MySQL, MariaDB, Microsoft Excel, PowerBI, Qlik, Looker Studio

Cloud Technologies: Google Firebase, AWS, Azure SQL, Google BigQuery

Certifications: Azure – AZ900, Google Cloud Crash Course, Data Science Math Skills - Duke University, IBM Data Science

EXPERIENCE

Data Science & Development Intern, iSimcha, LLC

May 2024 – Present

- Spearheaded the development of a **Python** process to efficiently extract and preprocess more than **500,000+** clinical trial records from nested JSON files, facilitating seamless integration into Firestore for data analysis.
- Conducted an assessment of the readability and comprehensibility of LLM-translated medical data using Python and **BigQuery** and created a comprehensive visualization dashboard for data-driven decision-making processes.
- Leveraged **ChatGPT, Mistral AI** and **Vertex AI's Gemini 1.0 Pro** Generative Model to transform complex clinical trial data into simplified terms in accordance with specific guidelines, enhancing readability for a wider audience.
- Integrated LLM processes with **Arize Phoenix LLM** and **Langfuse's LLM as a Judge** evaluation tools to enhance model performance monitoring, evaluation, and debugging, resulting in improved accuracy.
- Developed responsive, data-driven web applications using **ReactJS, CSS3**, and integrated with **Firebase** for efficient real-time data management.
- Implemented **Firebase Authentication** and anonymous user tracking for enhanced user engagement and analysis.

Senior Data Analyst, AXA XL

Feb 2022 – Jul 2023

- Designed and developed restructuring tool and new internal allocation process for insurance treaties which reduced calculation time by **40%**. Created visualization dashboards using **PowerBI**.
- Implemented new data analysis and automation tools using **SQL** and **Python** and optimized the allocation and restructuring process to improve efficiency and accuracy in actuarial processes.
- Built an integrated **Python** and **JavaScript** based data analysis and quarterly BAU automation tool.
- Created allocation and automation tool for actuarial modules, cutting time required to construct tools by **75%**.

PHP Web App Developer, Provin Technos Pvt. Ltd.

May 2020 – Aug 2020

- Built a web based **ERP system** and created user interface using **HTML5** and **CSS3** practices and used programming capabilities in **PHP, SQL** to create web application components and deployed using AWS.

PROJECTS

- **Computer Vision:** Developed **Python** based software for **Enabling Geriatric Exercising for Senior Population in Remote Areas** under TIH IoT CHANAKYA Graduate Grant offered by **IIT - Bombay**
- **Neural Networks:** Built an **upgraded CNN based model** for identifying and classifying Plant Diseases.
- **Machine Learning:** Created a comprehensive system to analyze gait factors, correlations and effects of speed and inclination with angles at various joints in the human leg using **Python** to understand **Gait Biomechanics**
- **NLP:** Explored sentiment in user-generated YELP Reviews using NLP techniques to gain insights into customers.
- **Database Management System:** Designed **Entity Relationship diagrams** and a database for a parking management system using **Java (JDBC), MySQL** and RDBMS **MariaDB**
- **Multi-agent Simulation:** Developed a multi-agent simulation using **Python's MESA** to analyze the manipulability of voting systems under varying levels of societal polarization, leveraging the Gibbard-Satterthwaite theorem.